SYSTEMATIC REVIEW PROTOCOL

Prepared in Accordance with PRISMA-P 2015 Guidelines

Artificial Intelligence for Brain-to-Speech Decoding:

A Systematic Review of Methods, Performance Benchmarks, and Generalization Across Neural Recording Modalities (2015 - 2026)

Protocol Type	Systematic Review Protocol (PRISMA-P 2015 Compliant)
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PROSPERO Reg. No.	
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Funding	No external funding. Institutional support only (non-commercial).

LATE REGISTRATION NOTICE

This protocol is submitted as a late registration. Preliminary work including literature scoping, protocol development, PICOS criteria definition, and preliminary database searches began in January 2025. The formal systematic database searches across all five databases will be executed on 13 April 206. Data extraction and formal screening had not commenced at the time of registration. The review methodology and eligibility criteria were fully specified prior to formal search execution, consistent with PRISMA-P guidance on late registration disclosure.

PRISMA-P 2015 Compliance Checklist

The table below maps each of the 17 mandatory PRISMA-P 2015 items to the section of this protocol where it is addressed.

Item	Section	Checklist Requirement	Addressed
1a	Title	Identified as protocol of a systematic review	**Yes**
1b	Title	Update of previous protocol: identified as such	**N/A**
2	Cover Page	PROSPERO registry name and registration number	**Pending**
3a	Cover Page	Author names, affiliations, email, mailing address	**Yes**
3b	Section 1	Author contributions and guarantor identified	**Yes**
4	Cover Page	Amendment plan / late registration disclosure	**Yes**
5a	Section 1	Funding sources indicated	**Yes**
5b	Section 1	Sponsor name provided	**Yes**
5c	Section 1	Role of funder/sponsor described	**Yes**
6	Section 2	Rationale described in context of existing knowledge	**Yes**
7	Section 3	Explicit research questions with PICO reference	**Yes**
8	Section 4	Full eligibility criteria: PICOS, study design, time frame, language	**Yes**
9	Section 5	All information sources with planned dates of coverage	**Yes**
10	Section 5	Full draft search strategy for at least one database	**Yes**
11a	Section 6	Data management mechanism described	**Yes**
11b	Section 6	Study selection process described through each phase	**Yes**
11c	Section 6	Data extraction method including piloting and investigator contact	**Yes**
12	Section 7	All data variables listed and defined	**Yes**
13	Section 7	All outcomes listed with prioritization and rationale	**Yes**
14	Section 8	Risk of bias methods described at study and outcome level	**Yes**
15a	Section 9	Criteria for quantitative synthesis described	**Yes**
15b	Section 9	Summary measures and methods for combining data	**Yes**
15c	Section 9	Additional analyses: subgroup analyses described	**Yes**
15d	Section 9	Type of summary if quantitative synthesis not appropriate	**Yes**
16	Section 9	Meta-bias assessment planned	**Yes**
17	Section 9	Certainty of evidence assessment method described	**Yes**

Section 1: Administrative Information

PRISMA-P Item 3b | Author Contributions and Guarantor

Blessing Isoyiza Adeika (Primary Reviewer / Guarantor): Conceived and designed the review, developed the PICOS framework and search strategy, conducted preliminary literature scoping, compiled and verified the included studies register, drafted the protocol, and will lead all stages of study selection, data extraction, quality assessment, and manuscript preparation.

Dr. Onyema Osuagwu (Supervisor / Co-author): Provided expert oversight of the review methodology, critically reviewed and approved the protocol, and will independently verify study selection and data extraction decisions. Acts as the secondary reviewer for all dual-review stages.

Blessing Isoyiza Adeika is the guarantor of this review, taking responsibility for the accuracy and integrity of the work as a whole.

PRISMA-P Item 4 | Amendments

This protocol represents a late registration and not an amendment of a previously published protocol. No prior version of this protocol has been registered on PROSPERO or any other systematic review registry. Any important amendments made to the protocol after registration including changes to eligibility criteria, search strategy, or analysis plan, will be documented with a rationale, dated, and submitted as amendments to the PROSPERO record. The audit trail will be publicly visible on the PROSPERO registration.

PRISMA-P Item 5a / 5b / 5c | Funding, Sponsor, and Role of Funder

This review receives no external, grant, or commercial funding. It is conducted as part of academic research within the Department of Electrical and Computer Engineering at Morgan State University. Institutional support is limited to access to library databases and supervisory time. No commercial funding or sponsorship is involved. Neither author has any conflict of interest related to the subject matter, methods, or conclusions of this review. The institution has no role in study design, data collection, analysis, interpretation, or the decision to publish.

Section 2: Introduction and Rationale

PRISMA-P Item 6 | Rationale

Brain-to-speech decoding the translation of neural signals into intelligible speech output using artificial intelligence has advanced rapidly over the past decade. Landmark invasive systems now achieve word error rates below 10% and communication rates approaching natural speech, representing potentially transformative technology for individuals with amyotrophic lateral sclerosis (ALS), locked-in syndrome, anarthria, and related conditions that cause severe or complete loss of speech. For this population, existing augmentative and alternative communication technologies remain slow, fatiguing, and inadequate substitutes for natural communication.

The primary engine of progress in this field is artificial intelligence. The trajectory runs from early linear decoders and support vector machines through convolutional and recurrent neural networks to the current era of transformer-based architectures, self-supervised learning, and large language model integration. Each transition has produced measurable improvements in decoding performance. However, the field faces a critical synthesis problem: despite rapid growth in the number of published studies, no prior systematic review has consolidated AI architecture evolution, quantitative performance benchmarks, and cross-participant generalization evidence across all major neural recording modalities in a unified PRISMA-compliant framework covering the period through to 2026.

Existing reviews are limited in one or more of the following ways: they focus on a single modality such as EEG or ECoG in isolation; they treat AI as a secondary technical detail rather than the primary subject of analysis; they do not apply a rigorous systematic methodology; or they predate the emergence of transformer-based and self-supervised approaches that now dominate the field. As a result, researchers entering the field lack a consolidated reference for which AI architectures perform best under which conditions, how performance metrics compare across modalities and task types, and which generalization strategies show genuine promise for scaling beyond the single-participant laboratory paradigm. Clinicians and translational engineers similarly lack a systematic basis for evaluating where the field stands relative to deployment readiness.

This review addresses that gap directly. It provides the field's first cross-modal benchmark reference covering 71 verified included studies across five modality groups, enabling systematic comparison of AI methods, performance, and generalization for the first time. A companion review by the same authors addresses trustworthiness dimensions interpretability, fairness, neuroprivacy, and AI governance which are deliberately excluded from this review to maintain focus and enable targeting of appropriate journals for each contribution.

Section 3: Objectives

PRISMA-P Item 7 | Research Questions (PICO-Referenced)

This systematic review addresses three pre-specified research questions. Each is anchored to the PICOS framework components:

RQ1 Methods

What artificial intelligence architectures and signal processing approaches (Intervention) have been applied to brain-to-speech decoding (Condition) across invasive and non-invasive neural recording modalities, in adults with speech/motor impairments or healthy participants (Population), and how have these approaches evolved from January 2015 to April 2026 compared to earlier methods (Comparator), as measured by architecture type, complexity, and adoption trends (Outcome)?

RQ2 Performance

What quantitative decoding performance benchmarks (Outcome) have been reported across neural recording modalities, decoding task types, and participant populations (Population) using AI-based decoding systems (Intervention), and are these benchmarks directly comparable across studies (Comparator) accounting for vocabulary size, task constraints, and evaluation methodology?

RQ3 Generalization

To what extent do current AI-based brain-to-speech decoders (Intervention) generalize across participants, recording sessions, languages, and recording modalities (Outcome), compared to single-participant, single-session systems (Comparator), and what methods have been proposed or demonstrated to address generalization gaps in adult populations with speech or motor impairments or healthy participants (Population)?

Section 4: Eligibility Criteria

PRISMA-P Item 8 | PICOS Eligibility Criteria

Population (P)

Inclusion:

- Human participants of any age, sex, or clinical status
- Participants undergoing neural or neuroimaging signal recording during any speech-related task, including overt speech, imagined or covert speech, attempted speech, perceived or listened speech, and silent articulation
- Clinical populations including individuals with ALS, locked-in syndrome, anarthria, brainstem stroke, spinal cord injury, and related conditions causing severe speech or motor impairment
- Healthy adult participants in proof-of-concept or non-invasive studies where neural speech signals are recorded and decoded

Exclusion:

- Animal studies, including non-human primate and rodent neural recording studies
- Studies conducted exclusively in pediatric populations where the neural decoding methodology is not generalizable to adult populations assessed case by case
- Studies involving no human neural or neuroimaging data

Intervention / Exposure (I)

The intervention is the application of AI-based computational decoding models to neural or neuroimaging signals to produce a speech-related output. Included model types:

- Convolutional neural networks (CNNs) and variants including DenseNet, ResNet, and EEGNet
- Recurrent neural networks including LSTMs and GRUs
- Encoder-decoder and sequence-to-sequence architectures
- Transformer and self-attention models
- Self-supervised and contrastive learning frameworks
- Large language model integration and brain-LLM alignment approaches
- Neural vocoder pipelines including WaveNet, HiFi-GAN, and BigVGAN
- Foundation model pre-training approaches for neural data
- Traditional machine learning classifiers including SVM and LDA were used as primary decoding model

Neural recording modalities covered:

- Invasive: Electrocorticography (ECoG), intracortical microelectrode arrays (Utah arrays), stereotactic EEG (sEEG), micro-ECoG (μECoG), fully implanted ECoG systems
- Non-invasive: Scalp EEG, magnetoencephalography (MEG), functional MRI (fMRI), functional near-infrared spectroscopy (fNIRS)
- Hybrid and peripheral: Combined EEG-fNIRS systems, high-density surface electromyography (HD-sEMG) for silent speech

Exclusion:

- Studies using computational models with no machine learning or AI component
- Studies decoding exclusively non-speech neural signals motor imagery, cursor control, P300 spellers where speech decoding is not a primary outcome

Comparators / Controls (C)

This review does not involve a single controlled comparator. Comparisons are structured along five pre-specified analytical dimensions:

- Modality comparisons: invasive versus non-invasive systems; ECoG versus intracortical versus EEG versus MEG versus fMRI
- Architecture comparisons: traditional ML versus deep learning; CNN versus RNN versus transformer versus self-supervised models
- Task comparisons: overt speech versus imagined or covert speech versus attempted speech versus perceived speech decoding
- Temporal comparisons: performance trajectories and architectural trends across the 2015 - 2026 period

- Generalization comparisons: single-participant versus cross-participant; single-session versus multi-session; within-language versus cross-language

Outcomes (O)

Primary: AI architecture type and evolution; quantitative decoding performance metrics (WER, classification accuracy, WPM, PCC, CER, BLEU/ROUGE, MOS); neural recording modality; generalization evidence; vocabulary scope.

Secondary: Participant characteristics; decoding task type; reproducibility indicators; study limitations; training data characteristics; language of decoding task.

Study Design (S)

Inclusion: Peer-reviewed empirical studies; conference papers from NeurIPS, ICLR, ICML, Interspeech, AAAI, and IEEE venues; preprints from arXiv and bioRxiv were substantive and quantitative labelled as preprints in synthesis.

Exclusion: Narrative reviews, systematic reviews, and meta-analyses; purely theoretical papers with no empirical human neural data; opinion pieces and editorials; conference abstracts without full methods and results; animal-only studies; studies without quantitative performance metrics; non-speech BCI studies.

Additional Restrictions

- Date range: January 2015 to April 2026
- Language: English only
- Publication status: Published and unpublished studies (including preprints) eligible

Section 5: Information Sources and Search Strategy

PRISMA-P Item 9 | Information Sources

Primary Electronic Databases

1. PubMed (MEDLINE)
2. IEEE Xplore
3. Scopus
4. Web of Science (Core Collection)
5. arXiv: cs.HC, eess.AS, and q-bio.NC subject categories

Supplementary Sources

- Backward citation tracing of all included studies and key prior reviews
- Forward citation tracking via Google Scholar for landmark papers: Willett et al. (2023); Moses et al. (2021); Tang et al. (2023); Défossez et al. (2023)
- Hand-searching of key journals: Nature, Nature Neuroscience, Nature Machine Intelligence, Journal of Neural Engineering, IEEE TNSRE, New England Journal of Medicine, Nature Communications, Current Biology, Neuron
- Screening of proceedings from NeurIPS, ICLR, ICML, Interspeech, and AAAI (2022 - 2026)
- Contact with authors or experts where recent unpublished or in-press work is identified

Search execution date: 13 April 2026

Date range: January 2015 - April 2026

Language restriction: English-language publications only

PRISMA-P Item 10 | Full Draft Search Strategy - PubMed

The following fully reproducible search string is provided for PubMed (MEDLINE). Database-specific adaptations for IEEE Xplore, Scopus, Web of Science, and arXiv will be documented in Appendix A of the final manuscript. The strategy uses three concept blocks combined with AND, with MeSH controlled vocabulary supplemented by free-text Title/Abstract searching.

Block 1: Neural Signal Acquisition

```
("brain-computer interface"[Title/Abstract] OR
"brain computer interface"[Title/Abstract] OR
"BCI"[Title/Abstract] OR
"brain-to-speech"[Title/Abstract] OR
"neural speech decoding"[Title/Abstract] OR
"speech neuroprosthesis"[Title/Abstract] OR
"neural speech prosthesis"[Title/Abstract] OR
"electrocorticography"[Title/Abstract] OR
"ECoG"[Title/Abstract] OR
"intracortical"[Title/Abstract] OR
"Utah array"[Title/Abstract] OR
"electroencephalography"[Title/Abstract] OR
"EEG"[Title/Abstract] OR
"magnetoencephalography"[Title/Abstract] OR
"MEG"[Title/Abstract] OR
"functional MRI"[Title/Abstract] OR
"fMRI"[Title/Abstract] OR
"fNIRS"[Title/Abstract] OR
"sEEG"[Title/Abstract] OR
"stereotactic EEG"[Title/Abstract] OR
"neural recording"[Title/Abstract] OR
"intracranial EEG"[Title/Abstract] OR
"iEEG"[Title/Abstract] OR
"Brain-Computer Interfaces"[MeSH Terms] OR
```

"Electroencephalography"[MeSH Terms] OR
"Magnetoencephalography"[MeSH Terms] OR
"Magnetic Resonance Imaging, Functional"[MeSH Terms])

Block 2: Speech Output

("speech decoding"[Title/Abstract] OR
"speech synthesis"[Title/Abstract] OR
"speech reconstruction"[Title/Abstract] OR
"speech restoration"[Title/Abstract] OR
"imagined speech"[Title/Abstract] OR
"covert speech"[Title/Abstract] OR
"inner speech"[Title/Abstract] OR
"silent speech"[Title/Abstract] OR
"attempted speech"[Title/Abstract] OR
"perceived speech"[Title/Abstract] OR
"speech neuroprosthesis"[Title/Abstract] OR
"augmentative and alternative communication"[Title/Abstract] OR
"AAC"[Title/Abstract] OR
"neural speech"[Title/Abstract] OR
"voice synthesis"[Title/Abstract] OR
"Speech"[MeSH Terms] OR
"Communication Aids for Disabled"[MeSH Terms] OR
"Speech Production Measurement"[MeSH Terms])

Block 3: Artificial Intelligence / Computational Methods

("deep learning"[Title/Abstract] OR
"machine learning"[Title/Abstract] OR
"neural network"[Title/Abstract] OR
"convolutional neural network"[Title/Abstract] OR
"CNN"[Title/Abstract] OR
"recurrent neural network"[Title/Abstract] OR
"LSTM"[Title/Abstract] OR
"long short-term memory"[Title/Abstract] OR
"transformer"[Title/Abstract] OR
"self-supervised learning"[Title/Abstract] OR
"transfer learning"[Title/Abstract] OR
"foundation model"[Title/Abstract] OR
"large language model"[Title/Abstract] OR
"LLM"[Title/Abstract] OR
"encoder-decoder"[Title/Abstract] OR
"sequence-to-sequence"[Title/Abstract] OR
"artificial intelligence"[Title/Abstract] OR
"deep neural network"[Title/Abstract] OR
"generative model"[Title/Abstract] OR
"self supervised"[Title/Abstract] OR
"pre-trained model"[Title/Abstract] OR
"Deep Learning"[MeSH Terms] OR
"Neural Networks, Computer"[MeSH Terms] OR
"Machine Learning"[MeSH Terms])

Filters Applied

AND "2015/01/01"[Date - Publication]: "2026/04/13"[Date - Publication]
AND "humans"[MeSH Terms]
AND "English"[Language]

Note: The three concept blocks are combined as: Block 1 AND Block 2 AND Block 3, then filtered by date, species, and language. Complete database-specific search strings for all five databases will be provided as Appendix A of the final published manuscript.

Section 6: Study Records

PRISMA-P Item 11a | Data Management

Records retrieved from all five databases will be exported and compiled in a reference management tool (Zotero or Mendeley) for deduplication prior to screening. Duplicate records will be identified using automated deduplication functions supplemented by manual review. A master screening log will be maintained documenting the disposition of every record included, excluded, or uncertain at each stage of the review, with reasons for exclusion recorded at full-text stage. The data extraction form and quality assessment forms will be maintained in a structured spreadsheet with version control. All review documents will be stored in a secure institutional repository with regular backups.

PRISMA-P Item 11b | Selection Process

Stage 1: Title and Abstract Screening

All records surviving deduplication will be independently screened by both reviewers against the eligibility criteria. Each record will be classified as Include, Exclude, or Uncertain. Records classified as Uncertain by either reviewer will automatically proceed to full-text assessment. Disagreements will be resolved through discussion and consensus.

Stage 2: Full-Text Assessment

Full texts of all included and uncertain records from Stage 1 will be obtained and independently assessed by both reviewers against the full PICOS eligibility criteria. Reasons for exclusion will be documented for every excluded full-text record. Disagreements will be resolved through discussion and, where necessary, by consultation with a third reviewer. Inter-reviewer agreement will be calculated using Cohen's kappa coefficient at both screening stages and reported in the final manuscript.

The complete screening and inclusion process will be reported using a PRISMA 2020 flow diagram in the final manuscript.

PRISMA-P Item 11c | Data Collection Process

A structured data extraction form was developed based on the review's PICOS framework and pre-specified outcomes. The form was piloted on five representative studies spanning each of the five modality groups prior to full extraction and refined to improve consistency and completeness. Data extraction will be performed by the primary reviewer. The secondary reviewer will independently cross-check all extracted data items. Discrepancies will be resolved through discussion until consensus is reached.

Where essential data required to answer the review's research questions are absent or ambiguous in published reports, study authors will be contacted via institutional email. A maximum of two contact attempts will be made. Non-responses will be documented and the study included with available data only, with the data gap noted in the synthesis.

Section 7: Data Items and Outcomes

PRISMA-P Item 12 | Data Items

The following variables will be extracted from each included study:

Study Metadata

- Authors and publication year
- Publication venue (journal, conference, preprint server)
- Digital Object Identifier (DOI)
- Publication status (peer-reviewed / preprint)

Neural Recording Modality

- Primary modality (ECoG, intracortical, sEEG, EEG, MEG, fMRI, fNIRS, sEMG, hybrid)
- Invasiveness category (invasive / minimally invasive / non-invasive)
- Electrode/sensor type and channel count were reported

Participant Characteristics

- Total participant count
- Clinical status (ALS, locked-in syndrome, anarthria, healthy, other)
- Age and sex where reported
- Ethics approval and consent procedures

AI Architecture

- Primary model architecture class (traditional ML, CNN, RNN/LSTM, encoder-decoder, transformer, self-supervised, LLM-integrated, hybrid)
- Input signal representation (raw signal, spectrotemporal features, high-gamma band, end-to-end)
- Training paradigm (supervised, self-supervised, semi-supervised, reinforcement learning)
- Language model integration (yes/no; type if yes)

Decoding Task

- Primary task type (overt speech, imagined/covert speech, attempted speech, perceived speech, silent articulation)
- Output type (text, phoneme labels, acoustic waveform, semantic representation)
- Vocabulary scope (closed-set / open-vocabulary)
- Vocabulary sizes were reported
- Language of decoding task

Performance Metrics

- All quantitative metrics reported: WER, classification accuracy (%), WPM, PCC, CER, BLEU, ROUGE, METEOR, MOS, listener intelligibility
- Evaluation methodology (held-out test set, cross-validation, leave-one-out)
- Whether results include confidence intervals or variance measures

Generalization Evidence

- Cross-participant generalization: yes/no; method; performance change
- Cross-session generalization: yes/no; method; performance change
- Cross-language generalization: yes/no; languages covered
- Cross-modal generalization: yes/no; modalities bridged

Reproducibility

- Source code availability (yes/no; repository link)
- Dataset availability (yes/no; access details)
- Pretrained model availability (yes/no)

Study Limitations

- Limitations explicitly stated by study authors

PRISMA-P Item 13 | Outcomes and Prioritization

Main Outcomes

1. AI architecture type and evolution: primary decoding model architecture classified by type; evolution across 2015–2026 mapped chronologically. Extracted from methods sections at time of publication.
2. Decoding performance: all quantitative metrics as reported, with priority - WER (open-vocabulary text decoding), classification accuracy (closed-set tasks), WPM (communication rate), PCC (acoustic reconstruction), CER (character-level), BLEU/ROUGE/METEOR (text generation), MOS and listener intelligibility (speech synthesis). Extracted at final reported evaluation timepoint.
3. Neural recording modality: classified into five groups - invasive ECoG/intracortical/sEEG; scalp EEG; MEG/fMRI; hybrid/sEMG; foundation model/transfer. Rationale: enables cross-modal benchmark comparison, the primary contribution of this review.
4. Generalization evidence: presence, type, method, and magnitude of cross-participant, cross-session, cross-language, or cross-modal generalization. Rationale: addresses RQ3, the most clinically critical gap in the field.
5. Vocabulary scope: closed-set versus open-vocabulary; vocabulary size where reported. Rationale: essential for benchmark comparability analysis in RQ2.

Additional Outcomes

1. Participant characteristics: sample size, clinical status, age, sex where reported
2. Decoding task: type overt, imagined/covert, attempted, or perceived speech
3. Reproducibility indicators: code, data, model availability
4. Study limitations: as explicitly stated by authors
5. Training data characteristics: dataset size, session count, recording duration
6. Language of decoding task: English or other; cross-language transfer evaluated

Section 8: Risk of Bias in Individual Studies

PRISMA-P Item 14 | Risk of Bias Assessment

Study quality will be assessed using an adapted Neural Decoding Study Quality Checklist informed by the CONSORT-AI extension for AI clinical trials and the TRIPOD+AI reporting guidelines. No established risk-of-bias tool exists specifically for AI-based neural decoding studies, which are neither randomized controlled trials nor purely diagnostic accuracy studies. The adapted checklist operationalizes quality assessment across seven dimensions relevant to this study type:

7. Reporting completeness - Are sufficient methodological details provided to reproduce the decoding pipeline? (Assessed at study level)
8. Participant representativeness - Is the participant sample described in sufficient detail, including clinical diagnosis, age, and sex? Are eligibility criteria stated? (Assessed at study level)
9. Performance metric appropriateness - Is the reported metric appropriate to the decoding task type? Is it reported with confidence intervals or variance measures? (Assessed at outcome level)
10. Evaluation rigor - Is performance evaluated on held-out test data unseen during training? Is cross-validation or test set separation explicitly described? (Assessed at outcome level)
11. Comparison baseline - Is performance compared against a meaningful baseline - chance level, prior state-of-the-art, or within-study ablation? (Assessed at outcome level)
12. Generalization testing - Is performance evaluated beyond a single participant and single session, or are the constraints of single-subject evaluation explicitly acknowledged? (Assessed at study level)
13. Reproducibility - Are code, data, or pretrained models publicly available? (Assessed at study level)

Each dimension will be rated as: Fully Met / Partially Met / Not Met / Not Applicable. Ratings will be applied at both the study level (items 1, 2, 6, 7) and outcome level (items 3, 4, 5) as specified above. Aggregate quality scores will be reported in a summary table and used to contextualize performance benchmark comparisons in the synthesis - studies rated as consistently Not Met across multiple dimensions will be flagged but not excluded, consistent with narrative synthesis guidance for technology evaluation reviews.

Quality assessment will be performed independently by both reviewers. Inter-reviewer agreement on quality ratings will be calculated using Cohen's kappa and reported in the final manuscript. Disagreements will be resolved through discussion and, where necessary, adjudication by a third reviewer. Where quality assessment is ambiguous due to incomplete reporting, authors will be contacted to clarify.

Section 9: Data Synthesis

PRISMA-P Item 15a | Criteria for Quantitative Synthesis

Quantitative meta-analytic synthesis will not be conducted. The primary outcome metrics such as Word Error Rate (WER), classification accuracy, Words Per Minute (WPM), Pearson Correlation Coefficient (PCC), and character error rate (CER) are defined and measured differently across included studies, and assessed on non-comparable vocabularies, participant populations, and task paradigms. Pooling these metrics would produce a misleading summary statistic that obscures rather than illuminates the evidence. This decision is consistent with established guidance on narrative synthesis for heterogeneous technology evaluation reviews (Popay et al., 2006; Campbell et al., 2020).

PRISMA-P Item 15b | Summary Measures

Where sufficient homogeneous subsets of studies exist for example, studies reporting WER on open-vocabulary intracortical decoding tasks, or studies reporting classification accuracy on closed-set EEG imagined speech tasks descriptive statistics will be reported: performance range (minimum to maximum), median value, and interquartile range where calculable. These descriptive summaries will be presented within the narrative synthesis to characterize the performance distribution within each subset, not as pooled effect estimates.

PRISMA-P Item 15c | Additional Analyses (Subgroup Analyses)

The following six pre-specified subgroup analyses will be conducted:

1. By modality group: Invasive (ECoG/intracortical/sEEG) versus non-invasive EEG versus MEG/fMRI versus hybrid/sEMG versus foundation model/transfer approaches
2. By decoding task: Overt speech versus imagined/covert speech versus attempted speech versus perceived speech
3. By vocabulary constraint: Closed-set (fixed vocabulary) versus open-vocabulary decoding
4. By participant type: Clinical populations (ALS, locked-in, anarthria) versus healthy participants
5. By temporal period: Pre-2020 versus 2020–2022 versus 2023–2026 to trace architectural evolution and performance trajectory
6. By generalization scope: Single-participant studies versus multi-participant or cross-participant studies

These subgroup analyses are designed to identify performance patterns and architectural trends that would be obscured by a global synthesis across all included studies simultaneously. They are pre-specified here to prevent post-hoc data dredging.

PRISMA-P Item 15d | Type of Summary - Narrative Synthesis

The primary synthesis method is narrative synthesis organized around the three research questions:

- RQ1 (Methods): AI architectures will be synthesized chronologically and by modality group, tracing evolution from traditional ML through CNN and RNN architectures to transformer-based and self-supervised approaches. An architecture evolution timeline figure will be produced mapping landmark papers to each transition point.
- RQ2 (Performance): Performance benchmarks will be synthesized in structured comparison tables stratified by modality group and decoding task type. A critical analysis of benchmark comparability will address vocabulary size effects, participant sample limitations, closed-set versus open-vocabulary paradigms, and language and dataset diversity gaps.
- RQ3 (Generalization): Generalization evidence will be synthesized by generalization type cross-participant, cross-session, cross-modal, cross-language documenting the methods used and the magnitude of performance changes attributable to each approach in a structured evidence table.

PRISMA-P Item 16 | Meta-Bias Assessment

Publication bias will be assessed at the review level. In brain-to-speech AI research, bias is expected to systematically favor studies reporting high decoding performance, novel architectures, or landmark clinical demonstrations over studies reporting negative results, failed generalization attempts, or modest improvements. This directional bias will be assessed through three complementary approaches:

1. Comparison of performance metrics across publication venues - peer-reviewed journal articles versus conference papers versus preprints, to identify systematic performance inflation in higher-prestige outlets

2. Examination of whether included studies explicitly report negative or null findings, failure cases, and generalization limitations alongside positive results
3. Narrative discussion of the likely direction and magnitude of publication bias in the review's limitations section, with explicit acknowledgment that the inclusion of preprints from arXiv and bioRxiv partially mitigates this bias by capturing results that have not undergone journal peer-review selection

Funnel plot analysis will not be conducted, as the heterogeneity of outcome metrics across studies precludes meaningful statistical assessment of publication bias.

PRISMA-P Item 17 | Certainty of Evidence

The certainty of findings will be assessed using an adapted GRADE (Grading of Recommendations Assessment, Development and Evaluation) approach. Standard GRADE domains will be applied and interpreted in the context of AI neural decoding research:

- Risk of bias: informed by the Neural Decoding Study Quality Checklist ratings from Section 8
- Inconsistency: assessed through examination of variability in performance metrics across studies within each modality and task subgroup
- Indirectness: assessed by examining the gap between study populations (often small, single-site, single-language cohorts) and the target clinical population
- Imprecision: assessed based on sample sizes, confidence intervals where reported, and vocabulary scope constraints
- Publication bias: informed by the meta-bias assessment in Section 9, Item 16

Given that most included studies involve single participants, small cohorts (median $n < 10$ in invasive studies), closed-set vocabularies, and non-standardized metrics, certainty ratings are anticipated to be low to very low across most outcome domains. This will be reported transparently in both the synthesis and the discussion as an honest characterization of the current evidence base's maturity - not as a limitation of the review methodology.

Section 10: Registration, Stage, and Dissemination

PROSPERO Registration

This protocol has been submitted for registration on PROSPERO, the international prospective register of systematic reviews maintained by the University of York Centre for Reviews and Dissemination. The PROSPERO registration number will be inserted in place of the placeholder on the cover page of this document once assigned. The registration number will also appear in the methodology section of the final published manuscript.

Stage of Review at Registration

Stage	Status at Registration
Pilot work and protocol development	**Completed (January 2025 – April 2026)**
Formal systematic database searching	**Scheduled - 13 April 2026**
Title and abstract screening	**Not yet started**
Full-text eligibility assessment	**Not yet started**
Data extraction	**Not yet started**
Risk of bias / quality assessment	**Not yet started**
Data synthesis and manuscript preparation	**Not yet started**

Related Reviews

Three prior reviews represent the closest existing work and are cited to justify the novelty of this review:

- Panachakel, J. T. & Ramakrishnan, A. G. (2021). Decoding covert speech from EEG - a comprehensive review. Frontiers in Neuroscience, 15, 642251. [Single modality - EEG only]
- Rabbani, H., Milsap, G. & Crone, N. E. (2019). The potential for a speech brain-computer interface using chronic electrocorticography. Neurotherapeutics, 16(1), 144–165. [Single modality - ECoG only]
- Lopez-Bernal, D., Balderas, D., Ponce, P. & Molina, A. (2022). A state-of-the-art review of EEG-based imagined speech decoding. Frontiers in Human Neuroscience, 16, 867281. [Single modality - EEG; single task - imagined speech]

A companion review by the same authors addressing trustworthiness dimensions of brain-to-speech AI - interpretability, fairness, neuroprivacy, and AI governance - will be submitted for separate PROSPERO registration. The two reviews address non-overlapping research questions.

Dissemination

The completed systematic review will be submitted to IEEE Transactions on Neural Systems and Rehabilitation Engineering (IEEE TNSRE, ISSN: 1534-4320) as the primary target journal, with Journal of Neural Engineering as a secondary target. A preprint will be posted on arXiv (cs.HC or eess.AS) prior to or concurrent with journal submission. The full included studies register, and data extraction tables will be made publicly available as supplementary material upon publication.

Keywords

Brain-computer interface; brain-to-speech decoding; neural speech neuroprosthesis; electrocorticography; ECoG; intracortical recording; electroencephalography; EEG; magnetoencephalography; MEG; functional MRI; fMRI; deep learning; transformer; convolutional neural network; self-supervised learning; transfer learning; foundation model; large language model; imagined speech; silent speech; speech synthesis; word error rate; generalization; systematic review; PRISMA; PRISMA-P; Morgan State University

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